

Joint All-Domain Command and Control

Joint All-Domain Command and Control, or **JADC2**, is a concept in active development by the United States Department of Defense to link sensors from all branches of the United States Armed Forces into a unified network powered by artificial intelligence.^{[1][2][3]}

Each branch of the United States Armed Forces has its own initiative that contributes directly to JADC2; the Army has *Project Convergence*,^[4] the Navy has *Project Overmatch*,^[5] the Air Force has the *Advanced Battle Management System* (ABMS),^{[6][7][8][9]} and the Space Force has the *National Defense Space Architecture* (NDSA) currently being developed by the Space Development Agency.^{[10][11][12]} See § *Outernet*

One primary application of JADC2 is a request—a call for fire (CFF).^[a] Combined JADC2 is almost ready for deployment pending Congressional approval of FY2024 funding.^{[13][14]}

Experimentation

The DoD has held at least two critical JADC2 exercises. The first demonstration of ABMS during an exercise in Florida in December 2019 centered on simulated threats posed by cruise missiles. Air Force and Navy aircraft (including F-22 and F-35 fighter jets), a Navy destroyer, an Army Sentinel radar system, a mobile artillery system, as well as commercial space and ground sensors, demonstrated their ability to collect, analyze, and share data in real-time to provide a more comprehensive picture of the operating environment.^[15] *For more information, see JADC2 at the Army's Project Convergence experiments*

In July 2020, the Department of Defense carried out a second test of the JADC2 system. In the course of this exercise, planes from the Air Force communicated with naval vessels that were stationed in the Black Sea. Additionally, special operations personnel from eight other NATO nations and a simulated environment collaborated to deter a possible attack from Russia.^[15]

In November 2022, ABMS experiments showed how JADC2 is really § combined arms. —Brig. Gen. Jeffery Valenzia (USAF CFT lead for JADC2) ^{[16][17]}

Infrastructure

In 2017, a joint network (Joint Enterprise Defense Infrastructure —JEDI) was proposed for DoD, with a single award meant for a single vendor for \$10 billion. Competitive bidding was held, and an award was made, but it was protested by a competitor. In 2021, the award was canceled;^[18] in its place, multiple vendors for an interoperable, compatible network capability with multiple awards were envisioned by 2022. This capability (Joint Warfighting Cloud Capability —JWCC) is to be for \$9 billion spread among

vendors and is meant to be awarded by mid-December 2022.^{[19][20]} Pentagon network officials began to envision JWCC as a necessary layer for JADC2.^[20] In early December 2022, JWCC was awarded to Google, Amazon, Microsoft, and Oracle.^{[21][22][23]}

DISA (Defense Information Systems Agency) was embarking on upgrading its JWICS (Joint Worldwide Intelligence Communications System), which is top-secret; DISA was using the same vendors as for JWCC, but also including IBM. Combining the JWICS with JWCC, economies of scale allowed a cooperative project of DISA's JWICS with DoD's JWCC.^[24] A Top Secret capability is being sought.^{[25][26]}

JADC2 Services

Decision superiority is an objective of CDAO.^{[27][28]} By FY2023, other Electromagnetic Battle Management System (EBMS) services will be available in JADC2.^[19] Raytheon BBN demonstrated its Robust Information Provisioning Layer (RIPL)^[29] which connects legacy links to ABMS.

A theater-level simulation tool will share data under a Cooperative Research and Development Agreement (CRADA) in order to visualize JADC2 scenarios.^[30]

The US Army's Integrated Tactical Network (ITN) will support JADC2 as a future capability. Integrated Tactical Network (ITN) Capability Set '25 will implement JADC2,^{[31][32]} according to the acting head of the Network CFT.^{[a][b]}

The Air Force's Common Tactical Edge Network joins 9 contractors, exploiting AI.^{[36][29]} Collins Aerospace demonstrated Combined JADC2 (CJADC2) in July 2022.^{[37][1][c][d]}

The Navy's Live, Virtual, and Constructive (LVC) environment is a training environment for naval operations, so that the Navy, Air Force, and Army can train together in a unified way.^{[38][39]}

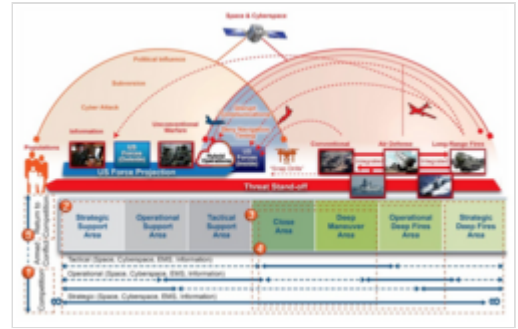
In Large Scale Exercise 2023 from August 9 to August 18, the Navy will test the response of over 50 Navy commands spread over 22 time zones, using retired Navy admirals to simulate members of the Joint Staff, civilian leaders, and non-Navy personnel.^{[40][41]} For example, aboard aircraft carrier USS Eisenhower (CVN-69), inside the Combat Information Center (CIC), the ship's watchstanding crew can train to the events presented to them on the screens of the CIC,^[42] irrespective of the physical position of USS Eisenhower during Large Scale Exercise 2023.^[42] The Navy's Project Overmatch is now installed on three carrier strike groups for § Project Convergence Capstone 4, or PCC4 (2024).^[43]

The following material is split from Army Futures Command

Multi-Domain Operations (MDO); Joint warfighting concept (JADC2)

- Multi-Domain Operations (MDO):^{[d][c][45][46][47][48]} Joint planning and operations are also part of the impending DoD emphasis on multi-domain operations.^{[49][50][51][52][53]} Multi-domain battalions,^[e] first stood up in 2019,^{[54][55]} comprise a single unit^{[56][57]} for air, land,^[58] space,^{[59][60][61][62]}—and cyber^{[63][64]} domains.^{[65][66][64]} A hypersonic-based battery similar to a THAAD battery is under consideration for this type of battalion,^{[67][68]} possibly denoted a *strategic fires* battalion^{[69][70][71]} (however, I2CEWS support would likely be

needed),^[e] depending on the theater. In 2019, these capabilities were analyzed as part of a series of globally integrated exercises.^{[72][73][74]} Using massive simulation,^{[47][75]} the need for a new kind of command and control (now denoted JADC2) to integrate this firepower was explored.^{[58][76]}



Multi-domain operations (MDO) span multiple domains: cislunar space, land, air, maritime, cyber, and populations.

- The ability to punch through any standoff defense of a near-peer competitor is the goal which Futures Command is seeking.^{[77][78][79][80]} For example, the combination of F-35-based targeting coordinates, long- range precision fires, and Low-earth-orbit satellite^[81] capability overmatches the competition, according to Lt. Gen. Wesley.^{[82][Note 2]} Critical decisions to meet this goal will be made based on data from the results of the Army's ongoing tests of the prototypes under development.^{[79][83]}
- For example, in Long Range Precision Fires (LRPF), the director of the LRPF CFT envisions one application as an anti-access/area denial (A2AD) probe; this spares resources from the other services;^[84] by firing a munition with a thousand-mile range at an adversary, LRPF would force an adversary to respond, which exposes the locations of its countermeasures and might even expose the location of an adversary force's headquarters. In that situation, an adversary's headquarters would not survive for long, and the adversary's forces would be subject to defeat in detail. But LRPF is only one part of the strategy of overmatch by a Combatant commander.
- In August–September 2020, at Yuma Proving Ground, the US Army engaged in a five-week exercise to rapidly merge multiple-domain capabilities. The exercise prototyped a ground tactical Network, pushing it to its limits of robustness^[85] (as of 2020, 36 miles on the ground, and demonstrated 1500-mile capability above the ground, with kill chains measured in seconds) in the effort to penetrate anti-access/area denial (A2AD) with long-range fires. Longer-range fires are under development, ranging from hundreds of miles to over 1,000 miles, with yearly iterations of Project Convergence being planned.^[86]
 - MDO (*multi-domain operations*) and JADC2 (*joint all-domain command and control*) thus entails:^[Note 2]
 1. Penetrate phase: satellites detect enemy shooters
 2. Dis-integrate phase: airborne assets remove enemy long-range fires
 3. Kinetic effect phase: Army shooters, using targeting data from aircraft and other sensors, fire on enemy targets.^[87]
 - Army Chief of Staff Gen. James C. McConville will discuss the combination of MDO^[c] and JADC2 with Air Force Chief of Staff Gen. Charles Q. Brown.^[85] In October 2020 the Chiefs agreed that Futures Command, and the Air Force's A5 office will lead a two-year collaboration 'at the most "basic levels" by defining mutual standards for data sharing and service interfacing' in the development of Combined Joint All-Domain Command and Control (CJADC2).^[88]
 - The ability of the joint services to send data from machine to machine was exercised in front of several of the Joint Chiefs of Staff in April 2021; this is a prerequisite capability for Convergence of MDO and JADC2.^{[89][90][91]}
- In July 2022 the 7th ASA(ALT) Doug Bush called for the formation of a large office on the scale of the Joint Counter-small UAS office, but for JADC2.^[92] This would coordinate,^[93] and eventually reconcile requirements for JADC2 for Army's Project Convergence, the

Navy's Project Overmatch and the Air Force's Advanced Battle Management System.^{[92][94][95][96]} See [CDAO](#)

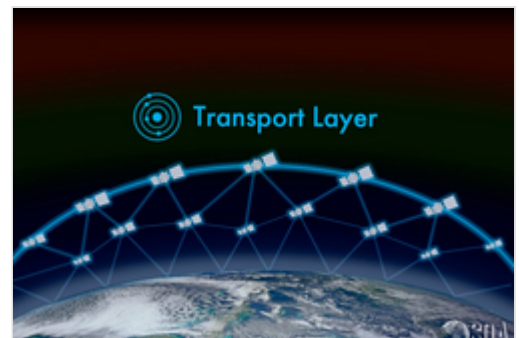
In April 2023, Joint Warfighting Concept 3.0 (JWC 3.0) was previewed.^[97] JWC 2.0 involved "fires, information, logistics, and command and control".^[97] Released 27 August 2023, JWC 3.0 additionally involves "the ability to win in contested logistics, information advantage, and expanded maneuver. — ADM Christopher Grady.^[98] Employment of crewed-uncrewed teaming to 'sense and make sense' of the situation, and to make that information rapidly available to the commanders will be an essential part of the Capability. Future Force Design is underway.^{[98][99]}

On 19 September 2023, Army Contracting Command-Aberdeen Proving Ground awarded [Leidos](#) a contract to provide an electronic platform, equipment and services in support of JADC2.^[100]

In February 2024, the Air Force's Brig. Gen. Luke Cropsey warned that phase two of Combined JADC2, which was supposed to go forward in 2024^[101] is not starting because Continuing Resolutions (CRs) do not fund new program spending —full Congressional budget approval for FY2024 is required for Combined JADC2 to go forward in 2024. From: [Transformation of the United States Army#cjadc2MeetIn60days](#)

Outernet

JADC2 confers on the US the capability to "move data globally at scale". —Gen. Chance Saltzman, US Space Force^{[104][105][106][107]} By year-end 2025, there will be 126 [Link-16](#) satellites operating in orbit, transmitting data in a worldwide [mesh network](#).^[108]



Transport layer of the National Defense Space Architecture (NDSA)^{[102] [103]}

Notes and References

- a. In 2009 the US Army replaced [command and control](#) with [mission command](#), in which the commander's intent suffices to empower subordinate unit commanders. This requires transparency of the command, and trust in the subordinates. For example, suppose the subordinate is a [squad leader](#);^[33] A forward observer (FO) in the squad requests fires (Call for fire -CFF); the CFF is a request, not an order. The fire direction center (FDC) reads back the data in 3 parts, as the forward observer sends the request:
 1. Observer identification and warning order.
 2. Target location.
 3. Description of target, method of engagement, and method of fire and control.
- b. [Call For Fire \(CFF\)](#)^{[34][35]}
- c. Echelons above brigade (division, corps, and theater army) engage in a [continuum of conflict](#).^{[Note 1][Note 2]}
- d. The Army's unclassified Multi-Domain Operations (MDO) concept is "the combined arms employment of capabilities from all domains that create and exploit relative advantages to

defeat enemy forces, achieve objectives and consolidate gains during competition, crisis, and armed conflict".^[44]

- e. When used in multi-domain operations, I2CEWS denotes *Intelligence, Information, Cyber, Electronic Warfare, and Space*. See: *ISR, or Intelligence, surveillance and reconnaissance*, and *National Geospatial-Intelligence Agency*
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 - "A computer-coordinated fight": in the air, land, sea, space, cyber, and the electromagnetic spectrum (EMS)
 - "forces from satellites to foot soldiers to submarines sharing battle data at machine-to-machine speed"
 - "it's the ability to integrate and effectively command and control all domains in a conflict or in a crisis seamlessly"—Gen. Hyten, Vice Chairman of the Joint Chiefs
 - All-Domain Operations (ADO) use global capabilities: "space, cyber, deterrent [the nuclear triad (for mutually assured destruction in the Cold War, an evolving concept in itself)], transportation, electromagnetic spectrum operations, missile defense".
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 - In September 2020, an ABMS Onramp demonstrated a specific scenario, which can be illustrated by the five red numbered bullet points from the slide in TRADOC pamphlet 525-3-1:
 1. Competition— No overt hostilities are yet detected. Blue bar (force projection) is in standoff against red bar (threat).
 2. Strategic Support area— National assets (blue) detect breaching of standoff by adversary (in red).
 3. Close area support— blue assets hand-off to the combatant commands, who are to create effects visible to the adversary (in red).
 4. Deep maneuver— blue combatant actions dis-integrate adversary efforts (per TRADOC pamphlet 525-3-1: "militarily compete, penetrate, dis-integrate, and exploit" the adversary); —Operational and Strategic deep fires create effects on the adversary. Adversary is further subject to defeat in detail, until adversaries perceive they are overmatched (no more red assets to expend).
 5. Adversary retreats to standoff. The populations perceive that the adversary is defeated, for now. (Compare to Perkins' cycle, 'return to competition', in which deterrence has succeeded in avoiding a total war, in favor of pushing an adversary back to standoff (the red threat bar). Blue force projection still has overmatched red threat.)
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